

A Study on H-1B Visa Applications and Their Denial Rates from 2017–2023

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March 20, 2026

Abstract

This study analyzes H-1B visa application trends and denial rates from 2017 to 2023, examining how policy changes, economic conditions, and industry dynamics influence approval outcomes.

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1 Abstract

The H-1B visa program enables U.S. employers to hire foreign professionals in specialized occupations when qualified domestic workers are unavailable. This study examines how political and economic disruptions influenced H-1B petition outcomes between 2017 and 2023. Using data from the USCIS H-1B Employer Data Hub, we analyze trends in visa approvals, denials, and petition volumes during a period shaped by the COVID-19 pandemic and a transition between presidential administrations with differing immigration policies. After cleaning and merging multi-year datasets, we compute longitudinal denial rates, analyze petition volume patterns, and construct a normalized metric of recruitment vitality across major industry sectors. Our findings show that denial rates increased significantly during the Trump administration before declining sharply following policy reversals under the Biden administration. In addition, the COVID-19 pandemic temporarily reduced overall petition volume and shifted employer behavior toward retaining existing workers rather than recruiting new foreign talent. These results highlight the sensitivity of H-1B outcomes to both policy interpretation and macroeconomic conditions.

2 Background

The H-1B program is designed for employers seeking to hire nonimmigrant visitors as workers in specialty occupations. A specialty occupation can be defined as one that requires specialized knowledge and the attainment of at least a bachelor’s degree or its equivalent. The H-1B is intended to help employers who cannot obtain needed skills from the U.S. workforce by authorizing the temporary employment of qualified individuals who are not otherwise eligible to work in the United States. In practice, however, H-1B sponsorship tends to be concentrated in certain industries. This concentration reflects the sectors where U.S. firms rely most heavily on highly trained foreign workers. The majority of authorized H-1B new hires nationally are employed in the Professional, Scientific, and Technical Services sector. Because these visas are closely tied to employment needs, broader economic conditions also influence visa demand and approval patterns. Despite the substantial slowdown in the growth of the U.S.-born labor force, immigration encourages economic expansion. This is because immigrants fill labor shortages while offsetting population aging since they tend to be younger and of working age. While long-term economic dynamics shape immigration trends, short-term shocks can disrupt them significantly. One of the most dramatic disruptions in recent history was the COVID-19 pandemic; it resulted in a major disruption to the economy beginning in early 2020. When the pandemic struck, economic activity plummeted. However, by June 2022, total employment had surpassed its pre-pandemic level. Although the overall economy recovered, the pace of recovery varied across industries. Professional and business services as well as technology sectors recovered quickly because of their ability to shift to remote formats, while sectors dependent on in-person interaction experienced the slowest recoveries. Notably, the industries most affected by COVID-19 were travel, transportation, and tourism. At the same time, immigration policy itself was

undergoing major changes. From 2017 to 2023, immigration policy shifted dramatically between presidential administrations, moving from the Trump administration’s “maximum pressure” approach between 2016–2020 to the Biden administration’s efforts to reverse many of these policies beginning in 2021. For the H-1B program specifically, the Trump administration increased scrutiny, which led to higher denial rates and a narrowing of key definitions. For example, the definition of “specialty occupation” was tightened to require that the bachelor’s degree be directly related to the position and that the position be so specialized that it could only be performed by an individual with a specific degree. Following this period of increased restrictions, the Biden administration attempted to reverse several of these policies. During Biden’s presidency, H-1B denial rates dropped to under 3%, and policy efforts shifted toward fair wages and worker protection. One change was the restoration of the “Prior Deference” policy, which instructed adjudicators to give weight to previous approvals when reviewing extension requests. These policy shifts, combined with the economic disruption of COVID-19, had significant implications for international students. The “Trump effect” increased visa denials and introduced discussions about eliminating Optional Practical Training (OPT), which reduced international student confidence in studying in the United States. At the same time, the pandemic sharply reduced new international enrollment and caused processing delays that hindered graduates’ ability to secure work authorization. The reduced ability for international students to remain in the United States resulted in fewer skilled workers available to U.S. firms, particularly in STEM fields. Between 2017 and 2020, regulatory changes forced many graduates to return to their home countries despite strong demand for their skills. However, during the Biden administration, Students benefited from several policy adjustments. Extensions to the F-1 student visa allowed graduates to transition more easily from student status to an H-1B

visa without employment gaps. Additionally, the definition of “specialty occupation” was clarified again so that while degree fields must relate to job duties, a broader range of qualifying fields could satisfy the requirement. As a result, international graduates entering the labor market between 2020 and 2023 faced an especially uncertain environment. They encountered an unstable labor market shaped by both visa-related policy changes and pandemic-related disruptions. Even though Optional Practical Training (OPT) remained an expanding pathway to employment, many highly skilled workers were still forced to leave the country due to shifting visa regulations and processing delays, making it difficult for employers to hire talent in critical fields such as STEM. Combining these shocks to the economy and changes to legislation has resulted in a highly uncertain environment for securing H-1B work authorization. Since the H-1B program sits at the intersection of labor market demand and immigration policy, changes to either can significantly affect the likelihood that foreign workers can work in the United States.

3 Introduction

The H-1B is a non-immigrant visa allowing U.S. employers to temporarily hire foreign professionals in “specialty occupations” requiring theoretical or technical expertise. This visa is essential to the U.S. labor market as it allows companies to recruit for the positions when they cannot find qualified U.S. workers. The probability of securing work authorization is highly susceptible to change, often fluctuating in response to economic downturns and shifts in political policy. The H-1B visa program provides a clear example of this dynamic, as approval rates and employer demand are closely tied to both the health of the U.S. economy and the direction of federal immigration policy. This project explores how two major disruptors, the COVID-19 pandemic and the transition between presidential

administrations, altered the probability of securing work authorization in the years 2017-2023. Beyond macroeconomic analysis, this research offers practical value to the university community as it identifies which industries remained welcoming for international talent during crises, providing guidance for the 12% of UCLA students navigating an increasingly complex immigration landscape.

4 Literature Review

During the Trump administration (2017–2021), several studies documented a sharp rise in H-1B denial rates driven by administrative policy changes rather than statutory reform. Research by the National Foundation for American Policy (2020) found that denial rates for initial H-1B employment petitions increased substantially between 2017 and 2019, particularly in specialty occupations, attributing this shift to heightened scrutiny and reinterpretation of eligibility standards. While this work establishes that approval probabilities became more restrictive under Trump, it largely focuses on aggregate trends rather than industry-specific disparities, leaving open the question of whether certain sectors, such as manufacturing, faced disproportionately higher denial rates during this period.

In 2020, during the period when COVID-19 was going rampant, and the Trump administration was taking office, it was reported that Trump signed a measure that would “prevent foreign workers from filling 525,000 jobs, according to White House estimates” (Miroff and Romm, 2020). This is reflected in one of our data visualizations, “Industry-Specific Denial Hurdles,” which reveals the drastic rise of denial rates in the six industries shown in the visual. Additionally, it wasn’t just preventing H1-B or foreign applicants from being able to obtain jobs in the U.S.; Trump also ordered a hiring freeze or “a pause” that affected

“more than half of the approximately 460,000 immigration visas issued last year” (Miroff and Romm, 2020). Due to these new restrictions, companies would be scrambling to find solutions for their hiring process involving H1-B applicants, and some companies had to choose to deny these applications to avoid attention from the government and/or delay the hiring process.

After the pandemic and Biden started to run the government administration, it was reported that the “relaxation of immigration policies of the Biden administration” (Chishti and Gelatt, 2022). This was in contrast to the “relaxation of immigration policies of the Trump administration” (Chishti and Gelatt, 2022). The Trump administration had implemented restrictive policies of the Trump administration” (Chishti and Gelatt, 2022). The Biden administration had implemented relaxation of immigration policies of the Biden administration” (Chishti and Gelatt, 2022).

5 Methods

5.1 Datasets

To evaluate H-1B visa trends amidst shifting political landscapes, we aggregated data from the USCIS H-1B Employer Data Hub spanning fiscal years 2017 through 2023. These datasets provide employer-level records of H-1B petitions, including counts of initial approvals, initial denials, continuing approvals, and other case outcomes, as well as associated industry classifications.

5.2 Exploratory Data Analysis

Exploratory data analysis (EDA) was conducted to examine the structure, distribution, and key patterns within the H-1B Employer Data Hub dataset. The dataset exhibits strong concentration effects across employers, industries, and geographic regions, motivating a deeper investigation into these dimensions.

To understand the distribution of H-1B activity across employers, we first aggregated initial petition counts by employer.

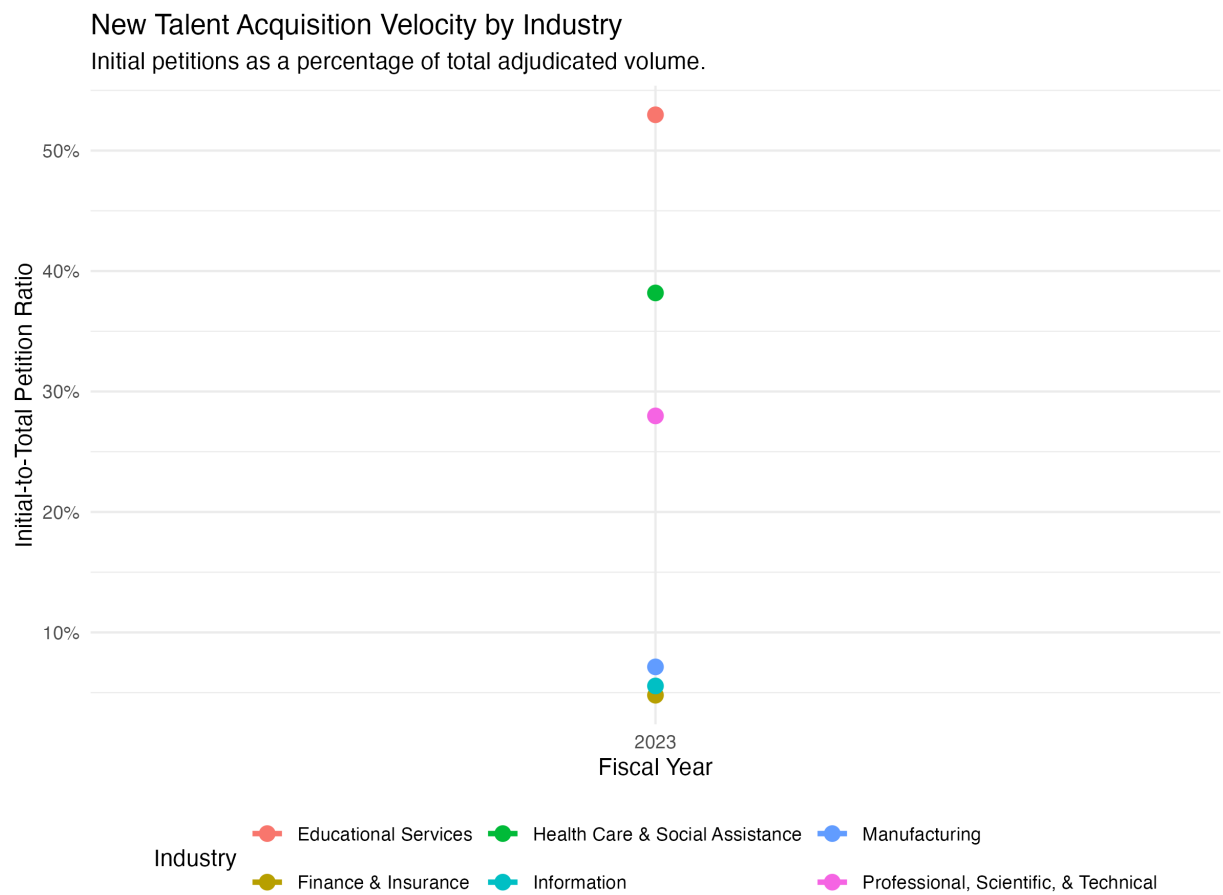


Figure 1: Top employers

Figure 1. Top 10 employers by total initial H-1B applications in 2023. The distribution is highly skewed, with a small number of firms—primarily large technology and consulting companies—accounting for a disproportionate share of total filings. This highlights the central role of major employers in driving H-1B demand.

This concentration suggests that the H-1B pipeline is not evenly distributed across firms, but instead dominated by a limited number of high-volume petitioners.

We next examined industry-level patterns by mapping NAICS codes to broader sector

categories and aggregating petition outcomes.

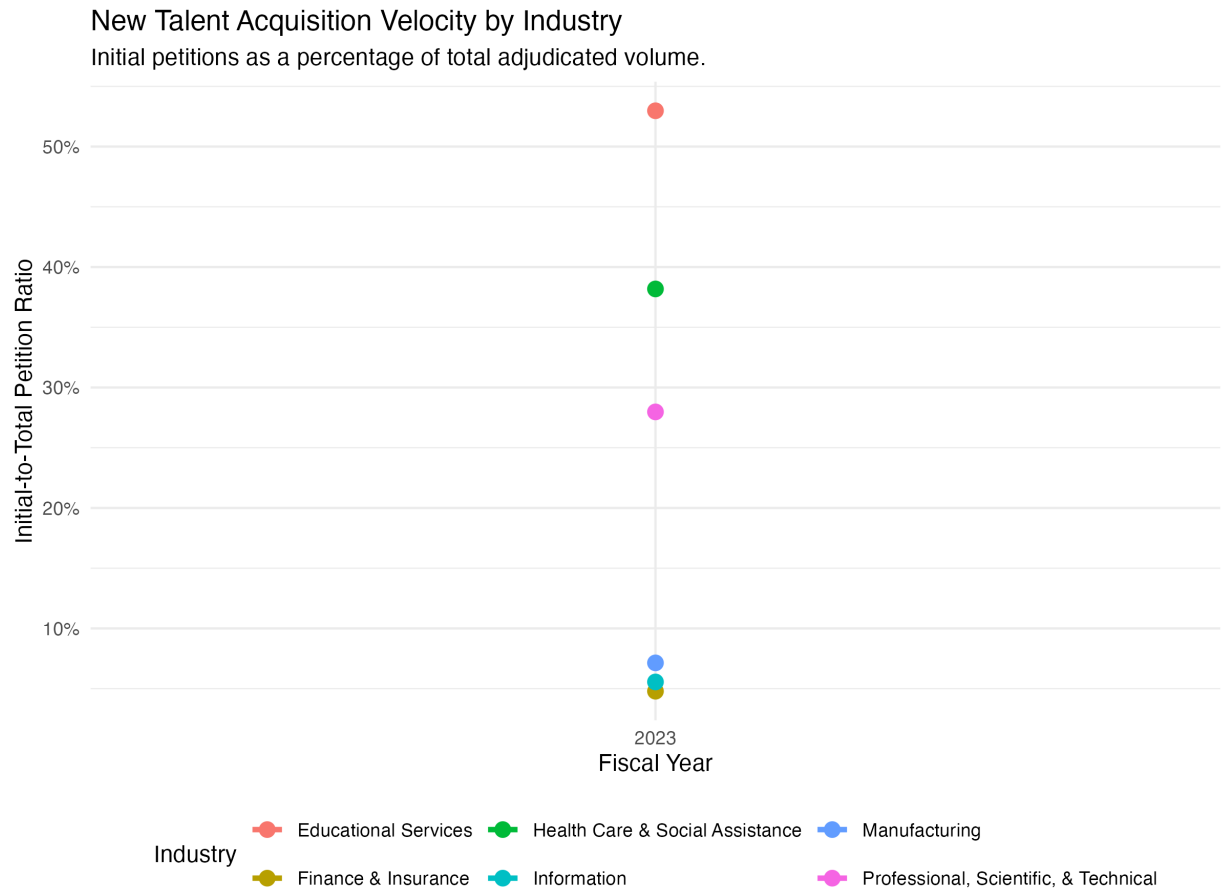


Figure 2: H-1B Volume by Industry

Figure 2. Total H-1B petition volume by industry (2023), broken down by approval and denial status. Professional, Scientific, and Technical Services dominates overall petition volume, followed by Manufacturing, Information, and Finance-related sectors.

Because absolute counts obscure differences in hiring behavior, we also examined the proportional composition of petitions within each industry.

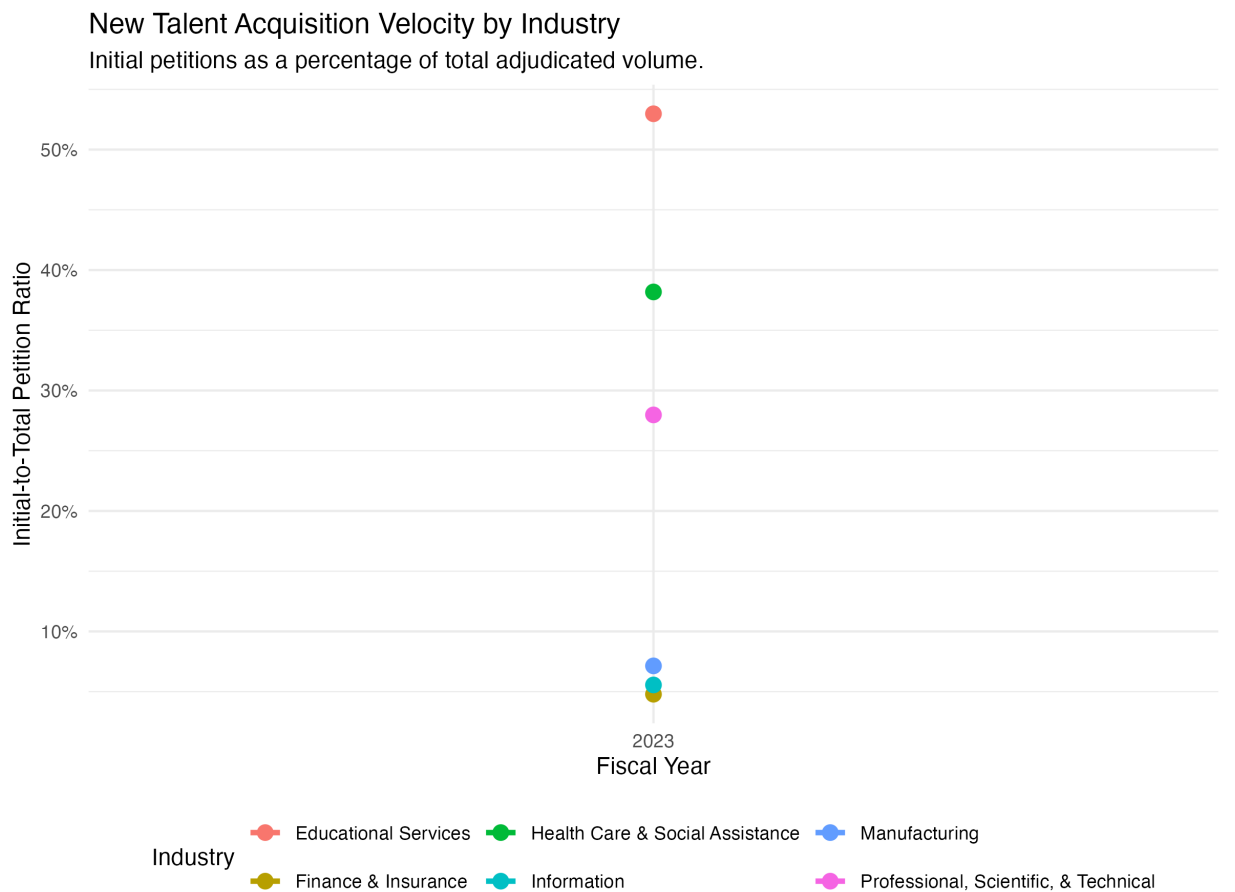
H-1B Composition by Industry

Figure 3. Proportional composition of H-1B petitions by industry (2023). While most industries are dominated by continuing approvals, sectors such as Health Care and Education

exhibit a relatively higher share of initial (new employment) petitions, indicating greater reliance on new foreign talent recruitment.

These results highlight important differences in hiring strategies across industries, with some sectors prioritizing workforce retention while others remain more active in recruiting new employees.

Geographic patterns were then explored by aggregating petition counts by state.



#need to make these plots smaller

Figure 4. Top 10 U.S. states by total H-1B petition volume in 2023. Petition activity is highly concentrated in states with strong technology and consulting sectors, such as California, Texas, and New York.

This geographic clustering reflects the concentration of industries that rely heavily on skilled

foreign labor.

Collectively, these visualizations reveal three key structural patterns: (1) a high concentration of H-1B activity among a small number of employers, (2) substantial variation in hiring strategies across industries, and (3) strong geographic clustering of petition demand. These insights informed the selection of the primary analytical metrics used in subsequent analysis.

6 Statistical Methods and Analytical Framework

6.1 Analytical Framework

To evaluate H-1B visa trends amidst shifting political landscapes, we aggregated USCIS H-1B Employer Data Hub records from fiscal years 2017 to 2023. Our methodology began with cleaning and merging these multi-year datasets to standardize inconsistent column definitions, followed by mapping North American Industry Classification System (NAICS) codes into distinct industry sectors to reduce noise.

We then employed longitudinal data aggregation and ratio normalization to assess shifts in petition outcomes across three primary dimensions. First, we calculated longitudinal denial rates, defined as initial denials divided by the sum of initial approvals and initial denials, to identify outcome volatility across different presidential administrations. Second, to analyze the petition volume and policy landscape, raw counts were categorized by case status, summed annually into a stacked temporal series, and overlaid with categorical shading to correlate volume fluctuations with specific administrative eras. Finally, to measure new talent acquisition velocity, we derived a relational metric by dividing initial petitions by

the total adjudicated volume across the six most active NAICS sectors. This normalization allowed for a comparative assessment of “recruitment vitality” across industries of disparate scales, independent of their absolute petition volumes.

6.2 Additional Analytical Techniques

Beyond these core components, several complementary statistical techniques were applied to enhance interpretability and support cross-sectional and longitudinal comparisons.

6.3 1. Industry Distribution Snapshot (2023)

This visualization tracks the annual probability of initial H-1B petition denial from 2017 to 2023, capturing how approval outcomes evolved over time. To construct this measure, yearly counts of initial approvals and initial denials were aggregated from the USCIS H-1B Employer Data Hub, and denial rates were computed as the ratio of initial denials to the total number of initial adjudications (initial approvals plus initial denials). These values were then plotted as a time-series line graph, with administrative periods annotated to contextualize policy shifts. The resulting trend reveals a sharp increase in denial rates between 2017 and 2020, followed by a rapid decline after 2021. This pattern suggests that changes in policy interpretation, rather than shifts in labor demand alone, had a substantial impact on visa outcomes. The elevated denial rates during the Trump administration and the subsequent decrease under the Biden administration indicate that administrative enforcement and adjudication standards play a critical role in determining access to H-1B work authorization.

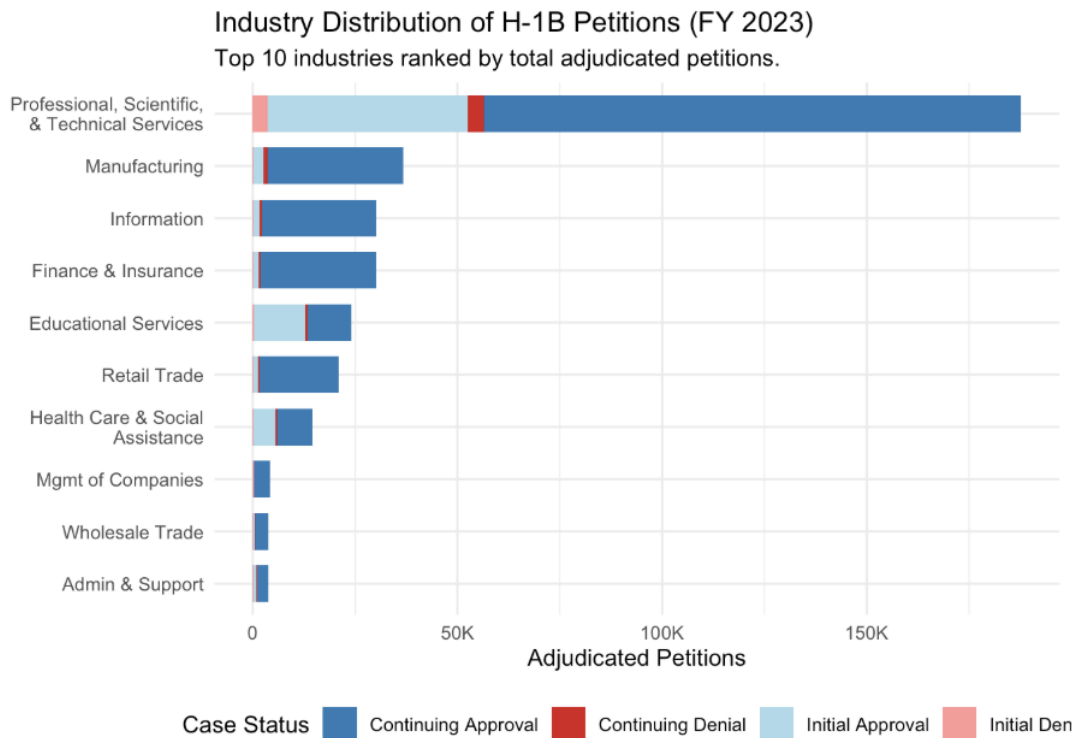


Figure 3: Industry distribution

This time-series framework allows us to identify shifts in outcome volatility across different administrative periods.

6.4 2. Sectoral Strategy: New Hires vs. Retention (2023)

This figure examines how industries allocate their H-1B petitions between new hires and continuing employees, providing insight into sector-level hiring strategies. Using the same aggregated 2023 dataset, petition counts were normalized within each industry by converting raw totals into proportions. This was implemented using `position = "fill"` in the bar chart, allowing each industry's bar to represent 100% of its activity regardless of scale. This normalization reveals structural differences in behavior: while high-volume industries such

as technology rely heavily on continuing petitions, sectors like education display a greater share of initial filings, indicating a stronger emphasis on recruiting new international talent.



Figure 4: New vs retention

6.5 3. Petition Volume & Policy Landscape (2017–2023)

This visualization tracks total H-1B petition volume over time, contextualized within shifting political administrations. The data were aggregated at the fiscal year level, summing all petition outcomes across industries to produce annual totals. The chart combines stacked bars (to represent petition composition) with background shading (implemented using `geom_rect()`) to distinguish between presidential administrations. This dual-layer design allows for simultaneous analysis of volume and policy context. The resulting trend highlights a clear contraction in petition volume during the COVID-19 period, followed by

recovery, illustrating how macroeconomic shocks and administrative environments jointly influence visa demand.

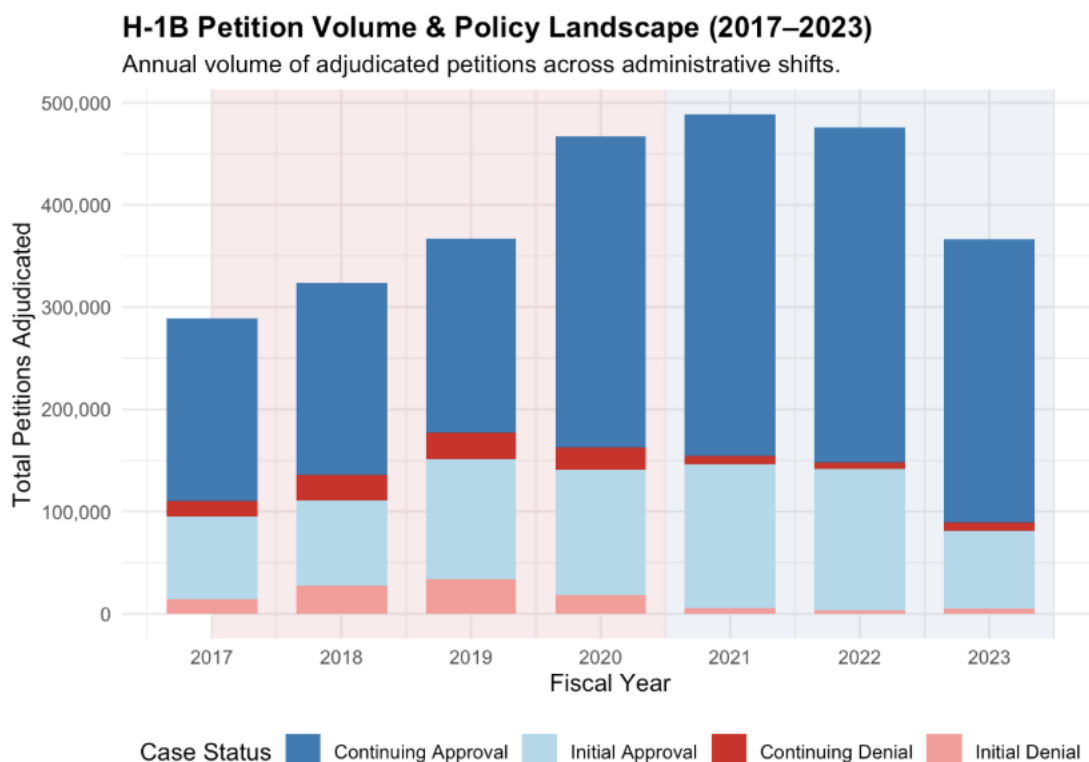


Figure 5: Petition volume and policy

6.6 4. Longitudinal Composition of H-1B Outcomes

This figure captures changes in the composition of petition outcomes over time, focusing on how the balance between initial and continuing petitions evolved. Using the same yearly aggregated dataset as the volume analysis, the data were transformed into proportions within each year using `position = "fill"`. This produces a stacked bar chart where each year sums to 100%, emphasizing relative composition rather than absolute scale. The visualization reveals a clear “pandemic pivot” in 2020, where the proportion of initial petitions declines while continuing petitions remain dominant, suggesting a systemic shift

toward workforce retention over new hiring during periods of uncertainty.

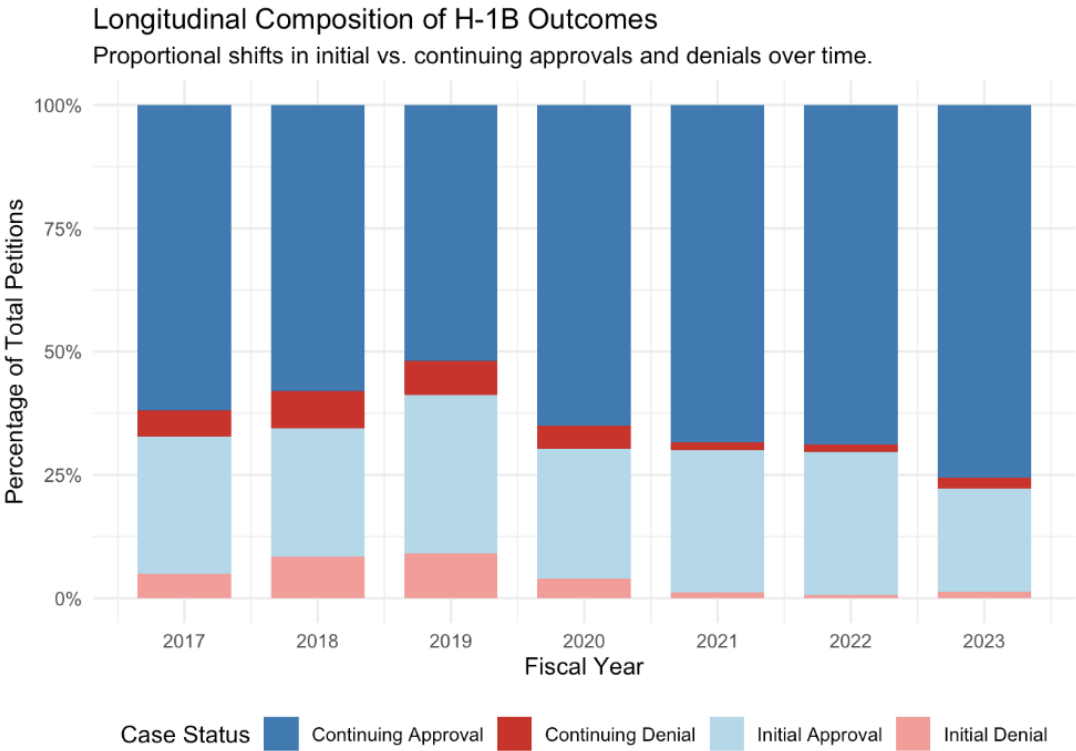


Figure 6: Longitudinal composition

6.7 5. New Talent Acquisition Velocity by Industry

This plot tracks the proportion of petitions allocated to new hires across key industries, offering a dynamic view of recruitment intensity over time. The metric was constructed by calculating the ratio of initial petitions to total petitions within each industry and year. To maintain interpretability, the analysis was restricted to the six most active industries. The visualization uses line plots with points (`geom_line()` and `geom_point()`), allowing trends to be compared across sectors. The results highlight divergent hiring strategies, with some industries maintaining stable recruitment pipelines while others shift toward

retention, reflecting varying responses to economic and regulatory uncertainty.

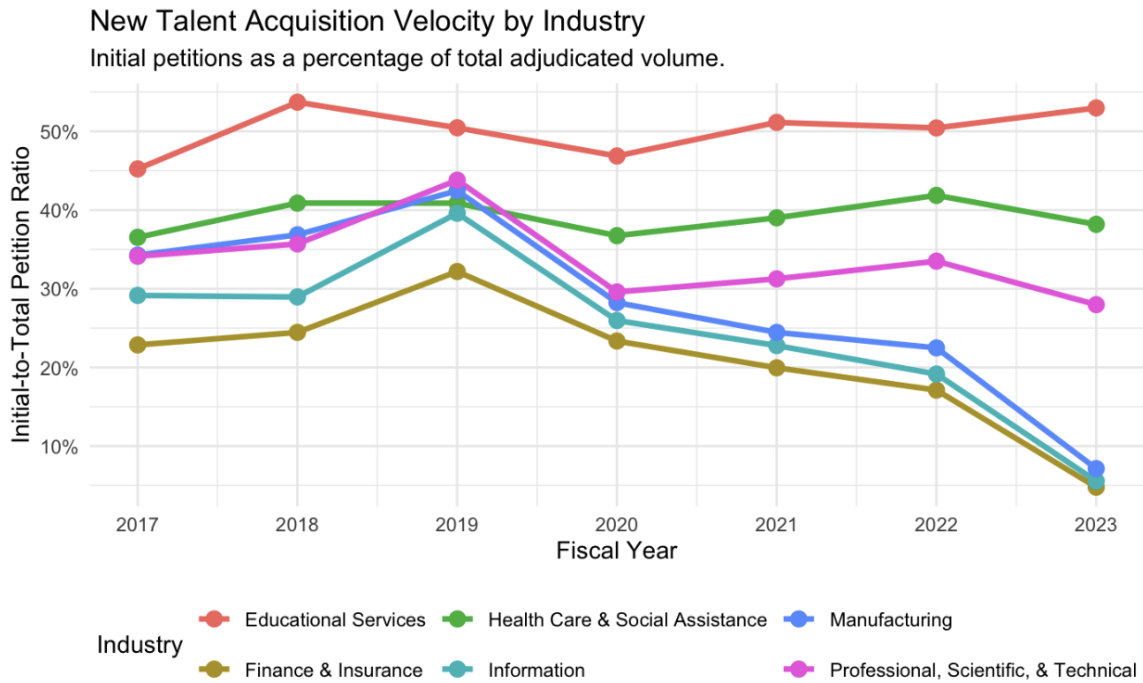


Figure 7: Talent acquisition velocity

6.8 6. Economic Resilience & Recovery Trajectories

This visualization examines how different industries responded to the economic disruption of 2020 by tracking their recovery trajectories over time. Petition volumes were aggregated by industry and year, and the results were visualized using faceted line charts (`facet_wrap()`). The use of `scales = "free_y"` allows each industry to have its own y-axis range, preserving the visibility of trends across sectors of vastly different sizes. This approach reveals distinct recovery patterns, with some industries exhibiting rapid “V-shaped” rebounds while others experience more gradual or delayed recovery, highlighting sector-specific resilience.

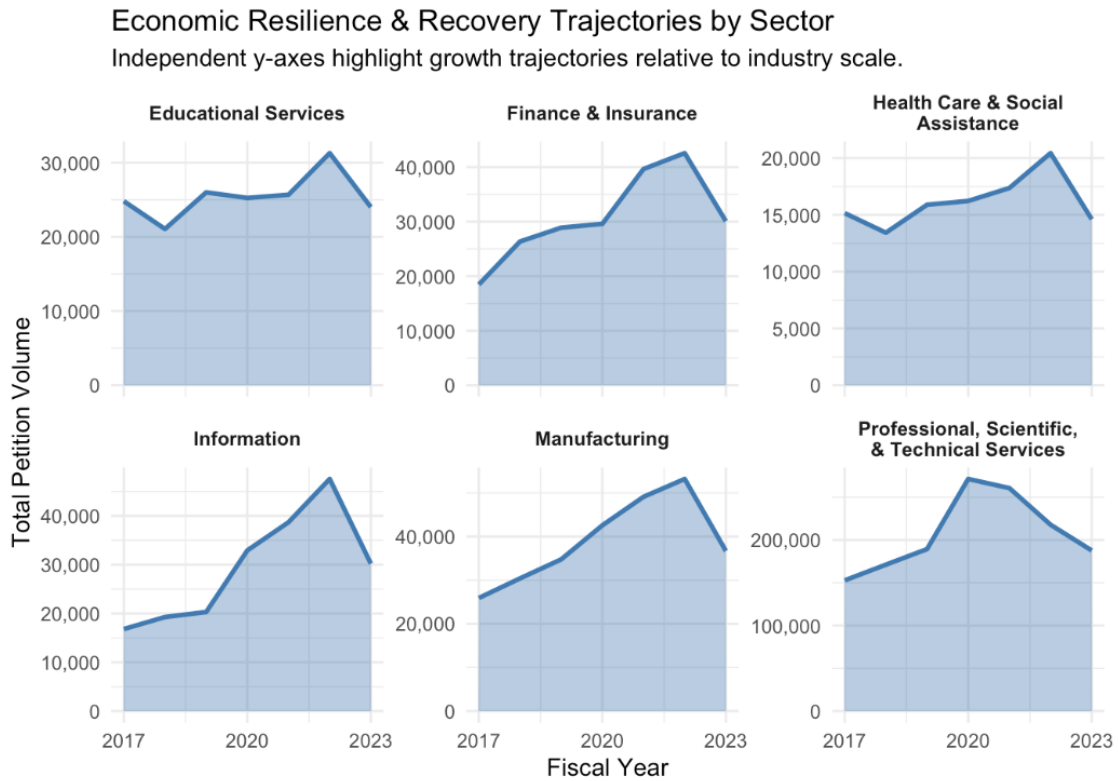


Figure 8: Economic resilience

6.9 7. Sector Profile: Professional Services (NAICS 54)

This figure provides a detailed examination of the largest sector within the H-1B program, Professional, Scientific, and Technical Services. The dataset was filtered specifically for NAICS 54, and petition counts were aggregated over time by outcome type. The visualization uses an area chart (`geom_area()`), which emphasizes the accumulation and flow of petition volume across years. This representation highlights how the sector's composition evolves and reinforces its role as a bellwether for the overall program, as trends within this dominant industry closely mirror broader H-1B dynamics.

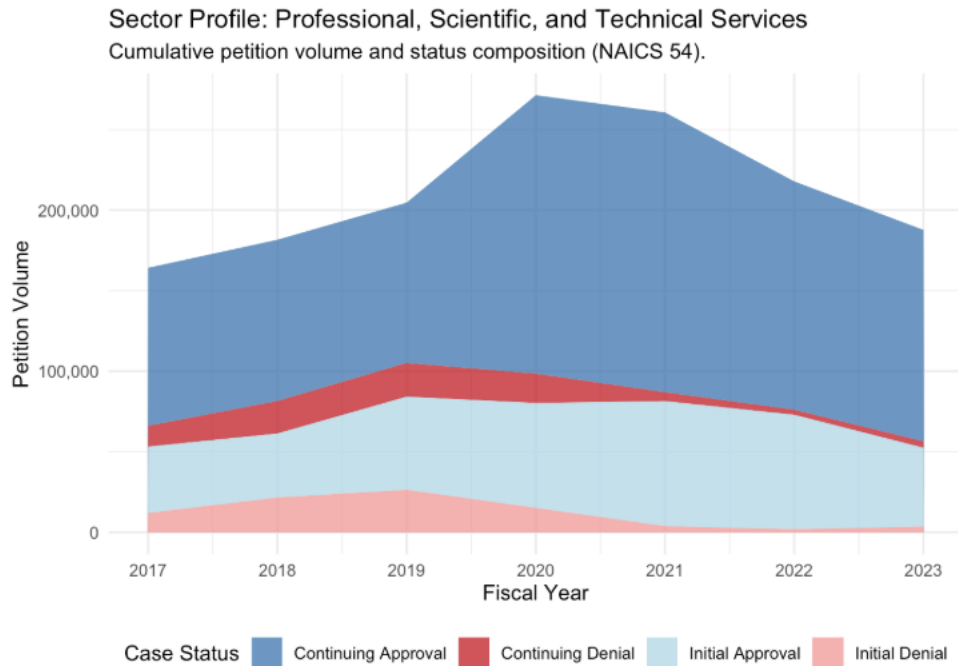


Figure 9: Sector profile

6.10 8. Consolidation Among High-Volume H-1B Sponsors

This analysis evaluates how concentrated H-1B activity is among the largest employers. For each year, employers were ranked by total petition volume, and the share of total approvals accounted for by the top entities was calculated. The resulting proportions were visualized using a bar chart. This method captures shifts in market concentration over time, revealing that a relatively small number of firms account for a disproportionate share of H-1B activity. Increases in concentration during certain periods suggest that access to the program may become more limited to large, established sponsors under specific economic or policy conditions.

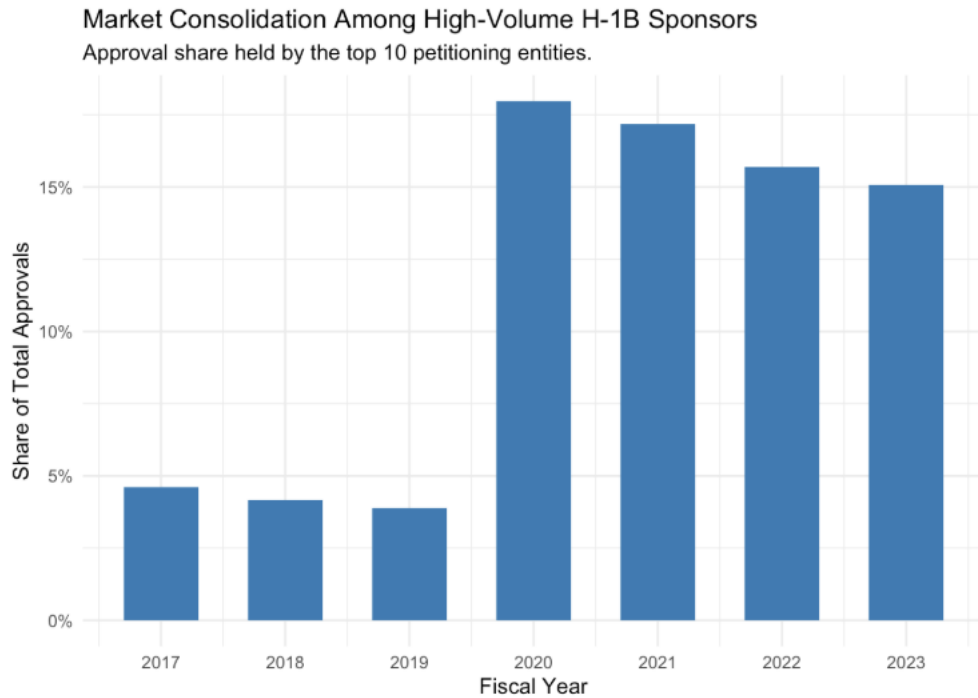


Figure 10: Market consolidation

6.11 9. Employer Size and Denial Rates

This figure compares denial rates across employers grouped by petition volume, providing insight into how firm size relates to approval outcomes. Employers were categorized into size bins based on total petition counts, and average denial rates were computed within each group. The visualization uses a bar chart to display differences across categories. The results suggest systematic variation in outcomes by employer size, with smaller firms often facing higher denial rates, potentially reflecting differences in legal resources, experience with the process, or petition quality.

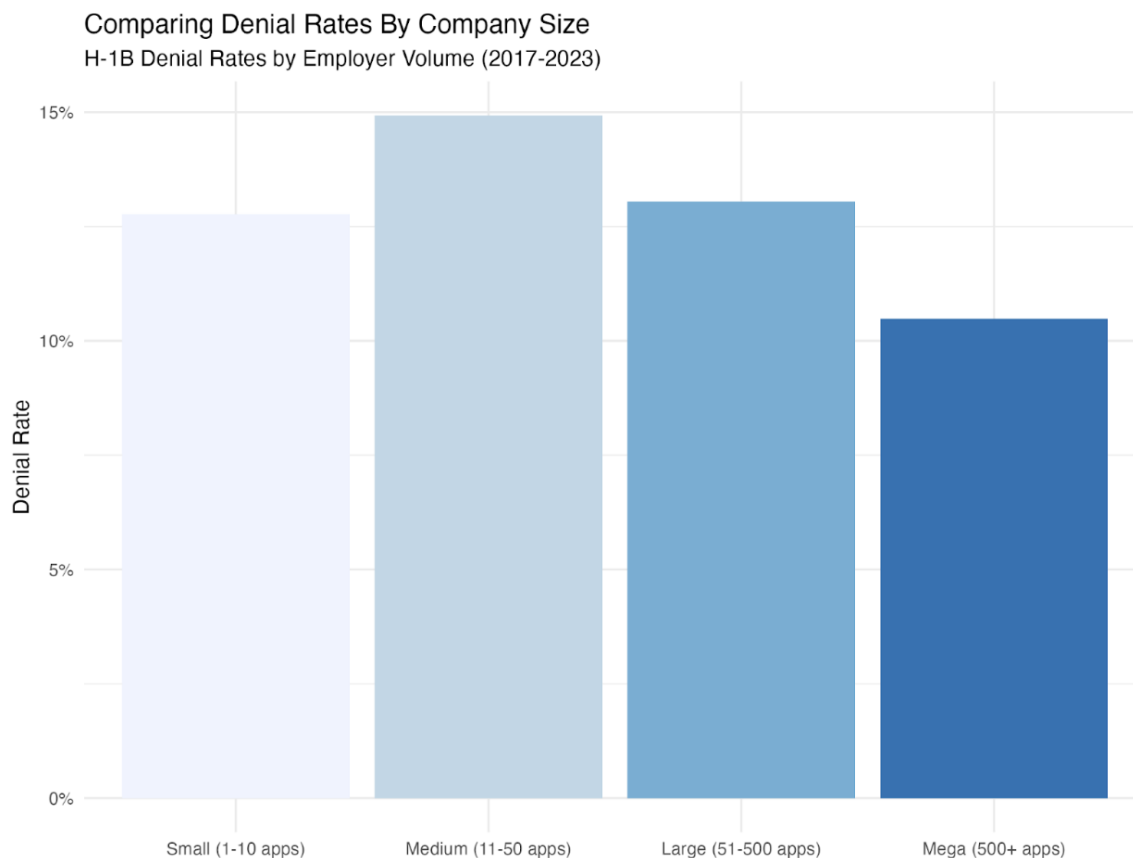


Figure 11: By company size

6.12 10. Industry-Level Denial Rates Over Time

These visualizations examine how denial rates vary across industries and over time. Denial rates were computed within each industry-year combination using the same ratio definition as the aggregate analysis. The first plot presents all industries together, while the second uses faceting to isolate trends within each sector. This dual approach allows for both cross-industry comparison and detailed inspection of individual trajectories. The results reveal that certain sectors, particularly professional and technical services, experienced sharper increases in denial rates during restrictive policy periods, indicating uneven impacts across

the labor market.

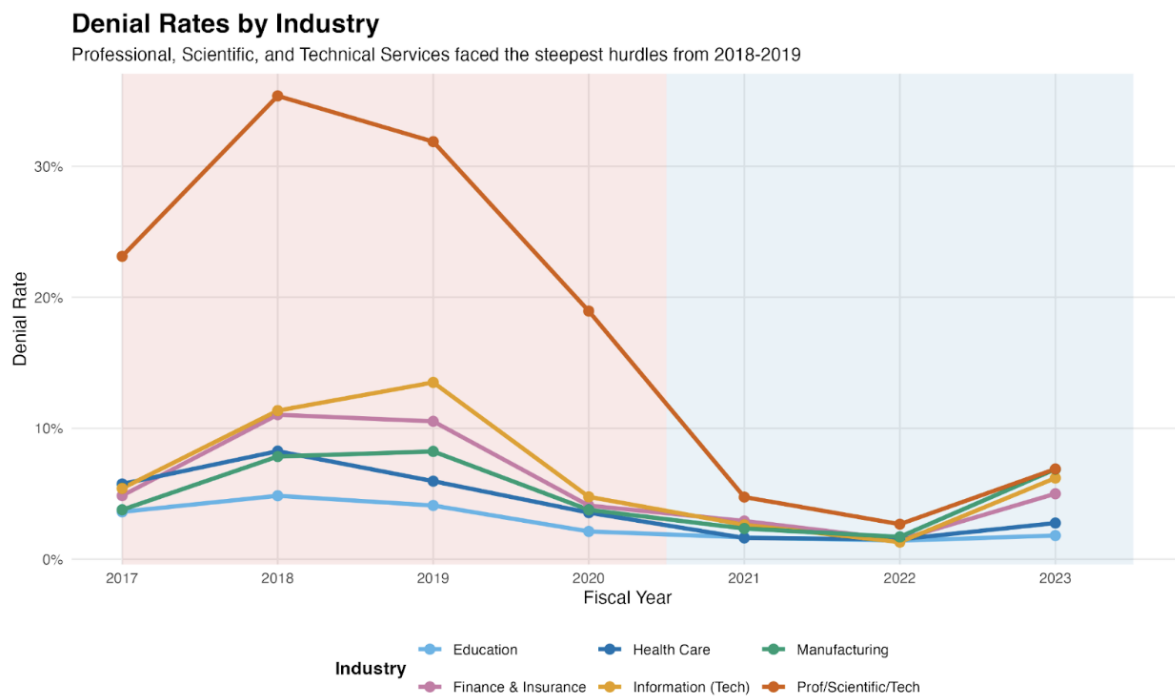


Figure 12: Denial rates by industry

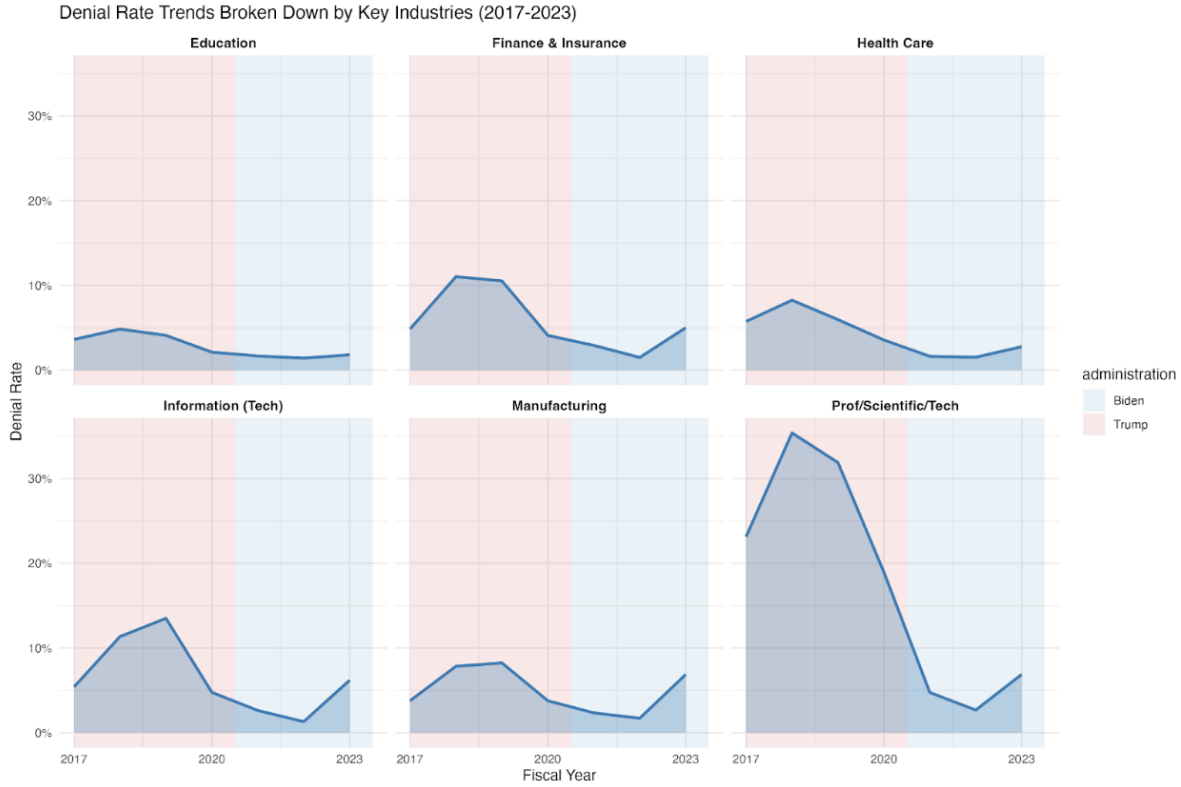


Figure 13: Industry breakdown

6.13 11. Heatmap of Industry Denial Rates

This heatmap visualizes denial rates across industries and years, providing a compact representation of temporal and sectoral variation. The data were structured into a matrix of industry-by-year denial rates and visualized using a color gradient, where darker shades indicate higher rejection probabilities. This format allows for rapid identification of patterns, such as concentrated periods of elevated denial rates and persistent differences across industries. The heatmap highlights clusters of high-denial environments during 2018–2019, reinforcing findings from the time-series analysis.

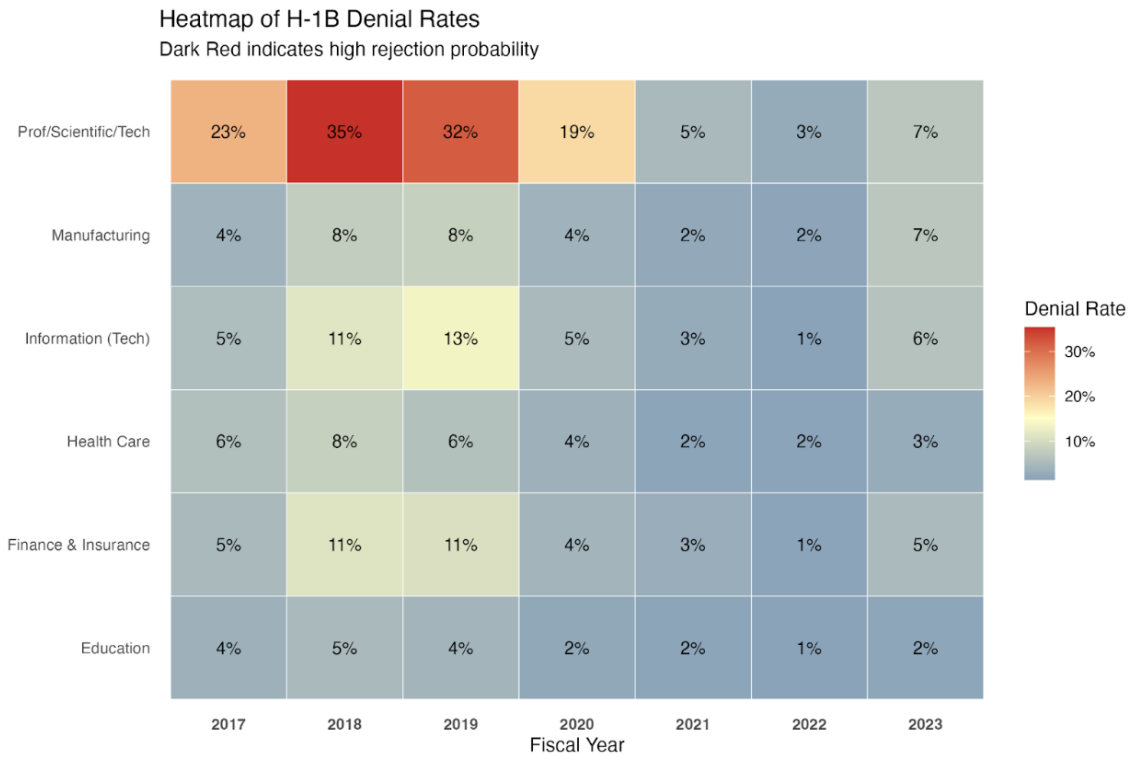


Figure 14: Heatmap

6.14 12. Geographic Distribution of Denial Rates

This visualization maps denial rates across U.S. states, illustrating regional disparities in H-1B outcomes. State-level denial rates were computed and merged with geographic boundary data to produce a choropleth map. The color gradient encodes rejection probability, allowing for spatial comparison. The results reveal that denial rates are not evenly distributed geographically, suggesting that regional factors such as industry composition or adjudication practices may influence outcomes.

Where is it Hardest to Get Approved?
Heatmap of Denial Rates by State (Min. 1,000 Apps, 2017-2023)

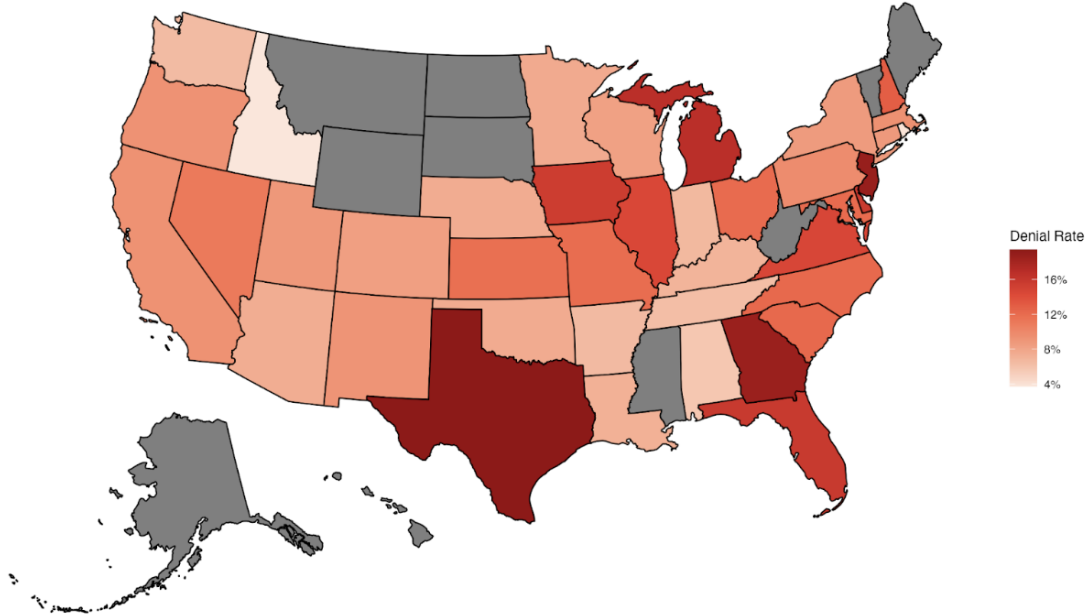


Figure 15: Geographic distribution

6.15 Summary

Together, these methods integrate data cleaning, aggregation, normalization, and time-series analysis to construct a comprehensive framework for evaluating H-1B visa dynamics. By combining absolute counts with proportional and ratio-based metrics, the analysis captures both scale and structure, enabling a nuanced understanding of how petition behavior varies across industries, firms, and political administrations.

7 Discussion

The results of this analysis illustrate how the probability of securing H-1B work authorization is shaped by forces beyond labor market demand alone. The longitudinal denial rate analysis reveals that policy interpretation under different presidential administrations had a measurable impact on visa outcomes. Denial rates increased sharply between 2017 and 2020, aligning with the Trump administration’s intensified scrutiny of H-1B petitions and reinterpretation of the “specialty occupation” requirement. This suggests that regulatory enforcement, even without statutory reform, can significantly alter the accessibility of work authorization for foreign professionals. At the same time, the petition volume analysis demonstrates how economic disruptions interact with policy changes. The COVID-19 pandemic produced a visible contraction in total petition volume as hiring slowed across industries. However, the magnitude of this contraction varied across sectors. Industries capable of remote work, particularly professional and technical services, recovered more rapidly than sectors dependent on physical presence. This divergence reinforces the importance of industry structure in shaping opportunities for foreign workers during economic shocks. The normalized “recruitment vitality” metric further highlights how industries differed in their willingness to bring in new international talent during periods of uncertainty. Even when total petition volume declined, some sectors maintained relatively stable proportions of initial petitions, indicating continued reliance on foreign talent pipelines. Other industries shifted toward extensions and retention of existing workers rather than recruiting new applicants. This shift suggests that during periods of regulatory or economic uncertainty, employers may prioritize maintaining their existing workforce over navigating the risks associated with new visa sponsorship. Finally, the geographic and industry-level patterns identified in the visualizations suggest that the impact of immigration policy changes

was not uniform across the country. Certain states and sectors continued to experience elevated denial rates even after federal policy changes began to reverse earlier restrictions. These disparities may reflect differences in adjudication practices, employer familiarity with the visa process, or regional industry composition. Taken together, these findings indicate that H-1B outcomes are shaped by the interaction of three major forces: immigration policy interpretation, macroeconomic conditions, and industry-level labor demand. The volatility observed between 2017 and 2023 highlights the sensitivity of the visa system to administrative priorities, reinforcing the idea that international professionals face a labor market where opportunities are influenced as much by policy context as by economic need.

8 Conclusion

This study examined how shifting presidential administrations and the COVID-19 pandemic reshaped H-1B visa outcomes across industries from 2017 to 2023. Together, the denial rate analysis and petition volume analysis paint a cohesive picture of a system highly sensitive to both political will and economic conditions. The findings confirm that H-1B visa outcomes were shaped less by shifts in labor demands than by the political environment. Denial rates climbed steeply during the Trump administration, peaking most severely before collapsing under Biden’s policy reversals. This volatility, driven by reinterpretation rather than statutory change, underscores how precarious work authorization can be for foreign-born professionals when eligibility standards shift with little warning. Layered on top of this political dynamic, the COVID-19 pandemic triggered a broad pivot away from new recruitment toward retaining existing workers. The recovery that followed was uneven across sectors, as industries capable of remote work regained momentum more quickly than others. Geographic disparities linked to political background further complicated the re-

covery, as certain states remained slow to reflect national-level policy improvements. The practical takeaway is that approval prospects depend heavily on industry choices and timing relative to political cycles, not qualification alone. As immigration policy enters another period of potential restriction, the patterns identified here offer a basis for anticipating where new barriers are likely to emerge, and which sectors have historically offered more stable pathways into the U.S. workforce.

9 Works Cited

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<https://www.pewresearch.org/short-reads/2025/03/04/what-we-know-about-the-us-h-1b-visa-program>.

Generative AI tools were used to assist in organizing the structure of the report, refining written explanations, and improving clarity of presentation. All data processing, statistical analysis, and visualization design were performed by the author. The use of AI was limited to editing support and did not replace independent analysis or interpretation.

10 R Code

Libraries

```
library(readr)
library(tidyverse)
library(janitor)
library(ggplot2)
library(scales)
library(ggrepel)
library(usmap)
```

Load in Data (2017-2023)

```
# need to load in 6 CSV files that match the pattern
file_list <- list.files(pattern = "h1b_datahubexport-.*\\.csv")

# read and combine them
h1b_all_years <- file_list |>
  map_dfr(~ read_csv(.x) |>
    clean_names() |>
    # 2017-2019 and 2020-2023 have some different col names,
    # so need to handle this
    rename_with(
      .fn = ~ str_remove(.x, "s$"),
      .cols = matches("approval|denial")
```

```

    ) |>

    type_convert()

  )

#glimpse(h1b_all_years)

#write_csv(h1b_all_years, "h1b-2017-2023.csv")

```

11 Denial Rate Dataframe

```

denial_rates <- h1b_all_years |>

  group_by(fiscal_year) |>

  summarise(

    total_initial_apps = sum(initial_approval + initial_denial, na.rm = TRUE),

    total_initial_denials = sum(initial_denial, na.rm = TRUE)

  ) |>

  mutate(

    denial_rate = total_initial_denials / total_initial_apps

  )

```

Denial Rate Across Years Shaded by Administration

```

# color dataframe to indicate administration

eras <- data.frame(

  administration = c("Trump", "Biden"),

  start = c(2017, 2020.5),

```

```

    end = c(2020.5, 2023.5)
  )

ggplot() +

  # add background shading

  geom_rect(data = eras,

            aes(xmin = start, xmax = end, ymin = -Inf, ymax = Inf, fill = administration),

            alpha = 0.1) +

  geom_line(data = denial_rates, aes(x = fiscal_year, y = denial_rate),

            color = "#8C8C8C", linewidth = 1) +

  geom_point(data = denial_rates, aes(x = fiscal_year, y = denial_rate),

            color = "black", size = 3) +

  # add labels

  geom_text_repel(data = denial_rates,

                  aes(x = fiscal_year, y = denial_rate,

                      label = percent(denial_rate, accuracy = 0.1)),

                  fontface = "bold",

                  size = 3,

                  bg.color = "white",

                  bg.r = 0.15,

                  force = 2,

                  segment.color = NA) +

  scale_fill_manual(values = c("Trump" = "#d7191c",

                              "Biden" = "#2c7bb6"),

```

```

      name = "Administration") +

scale_y_continuous(labels = percent_format(accuracy = 1), limits = c(0, 0.3)) +

scale_x_continuous(breaks = 2017:2023) +

labs(

  title = "The 'Trump Effect' on H-1B Visas",

  subtitle = "Initial Denial Rates by Fiscal Year (2017-2023)",

  x = "Fiscal Year",

  y = "Denial Rate",

  caption = "Source: USCIS H-1B Employer Data Hub"

) +

theme_minimal() +

theme(

  plot.title = element_text(face = "bold", size = 16),

  panel.grid.minor = element_blank(),

  legend.position = "right",

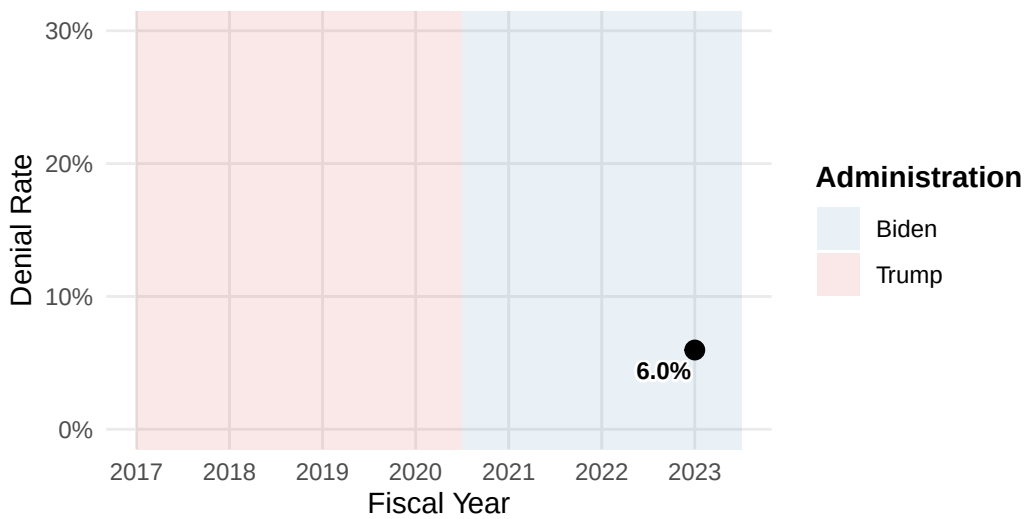
  legend.title = element_text(face = "bold")

)

```

The 'Trump Effect' on H-1B Visas

Initial Denial Rates by Fiscal Year (2017–2023)



Source: USCIS H-1B Employer Data Hub

```
#ggsave("denial_rate_time_series.png", width = 8, height = 5, bg = "white")
```

Industry Analysis

```
naics_map <- c(
  "11" = "Agriculture", "21" = "Mining/Oil/Gas", "22" = "Utilities",
  "23" = "Construction", "31" = "Manufacturing", "32" = "Manufacturing", "33" = "Manufa",
  "42" = "Wholesale Trade", "44" = "Retail Trade", "45" = "Retail Trade",
  "48" = "Transportation", "49" = "Transportation",
  "51" = "Information (Tech)", "52" = "Finance & Insurance",
  "53" = "Real Estate", "54" = "Prof/Scientific/Tech",
  "55" = "Management", "56" = "Admin & Support",
  "61" = "Education", "62" = "Health Care",
  "71" = "Arts/Entertainment", "72" = "Accommodation/Food",
  "81" = "Other Services", "92" = "Public Admin"
```

```

)

top_industries_list <- h1b_all_years |>

  # ensure NAICS is a character for matching
  mutate(naics_code = as.character(naics)) |>

  # map codes to names
  mutate(industry = naics_map[naics_code]) |>

  # handle unmapped/missing industries
  mutate(industry = ifelse(is.na(industry), "Other/Unknown", industry)) |>

  group_by(industry) |>

  summarise(

    grand_total_apps = sum(initial_approval + initial_denial, na.rm = TRUE)

  ) |>

  arrange(desc(grand_total_apps)) |>

  slice_head(n = 6) |>

  pull(industry) # extract just the names as a list

print(top_industries_list)

```

Denial Rates by Industry

```

industry_trends <- h1b_all_years |>

  mutate(naics_code = as.character(naics)) |>

  mutate(industry = naics_map[naics_code]) |>

  mutate(industry = ifelse(is.na(industry), "Other/Unknown", industry)) |>

```

```

# group and calculate

group_by(fiscal_year, industry) |>
summarise(

  total_apps = sum(initial_approval + initial_denial, na.rm = TRUE),

  denials = sum(initial_denial, na.rm = TRUE),

  .groups = "drop"

) |>

# calculate Rate

mutate(denial_rate = denials / total_apps) |>

# IMPORTANT: Filter for only the "Big 6" industries to keep plots readable
# (You can adjust this threshold, but these usually make up 80%+ of volume)

filter(industry %in% top_industries_list)

color_str <- c(

  "Prof/Scientific/Tech" = "#D55E00", "Information (Tech)" = "#E69F00",

  "Finance & Insurance" = "#CC79A7", "Manufacturing" = "#009E73",

  "Health Care" = "#0072B2", "Education" = "#56B4E9"

)

ggplot(industry_trends, aes(x = fiscal_year, y = denial_rate, color = industry)) +

  # background shading

  geom_rect(data = eras, inherit.aes = FALSE,

            aes(xmin = start, xmax = end, ymin = -Inf, ymax = Inf, fill = administration

              alpha = 0.1) +

```

```

geom_line(linewidth = 1.2) +

geom_point(size = 2.5) +

scale_y_continuous(labels = percent_format(accuracy = 1)) +

scale_x_continuous(breaks = 2017:2023) +

scale_color_manual(values = color_str) +

scale_fill_manual(values = c("Trump" = "#d7191c", "Biden" = "#2c7bb6"),
                    guide = "none") +

labs(

  title = "Denial Rates by Industry",

  subtitle = "Professional, Scientific, and Technical Services faced the steepest hurt",

  x = "Fiscal Year",

  y = "Denial Rate",

  color = "Industry"

) +

theme_minimal() +

theme(

  legend.position = "bottom",

  plot.title = element_text(face = "bold", size = 16),

  legend.title = element_text(face = "bold"),

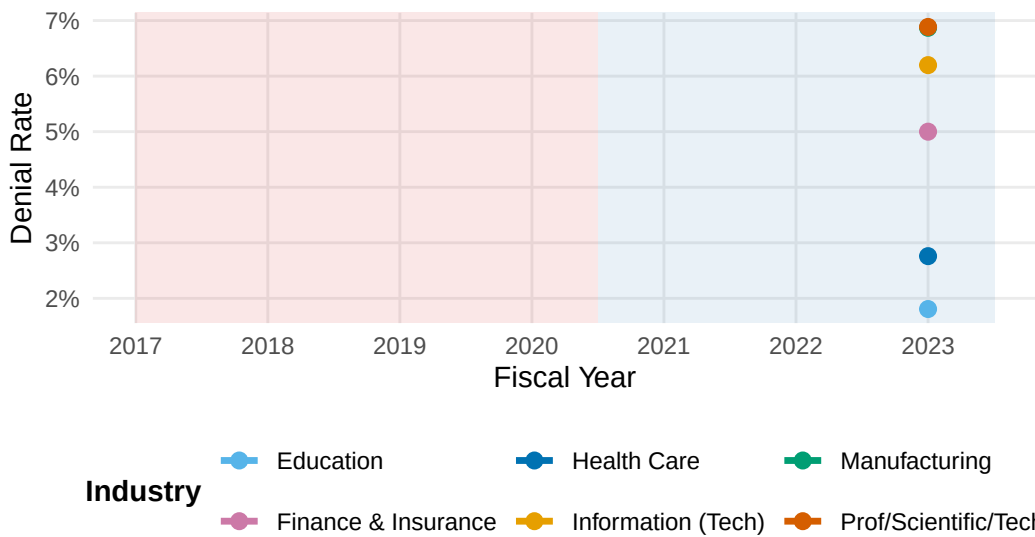
  panel.grid.minor = element_blank()

)

```

Denial Rates by Industry

Professional, Scientific, and Technical Services faced the steepest hurdles



```
ggsave("denial_rate_industry_lines.png", width = 10, height = 6, bg = "white")
```

Heat Map

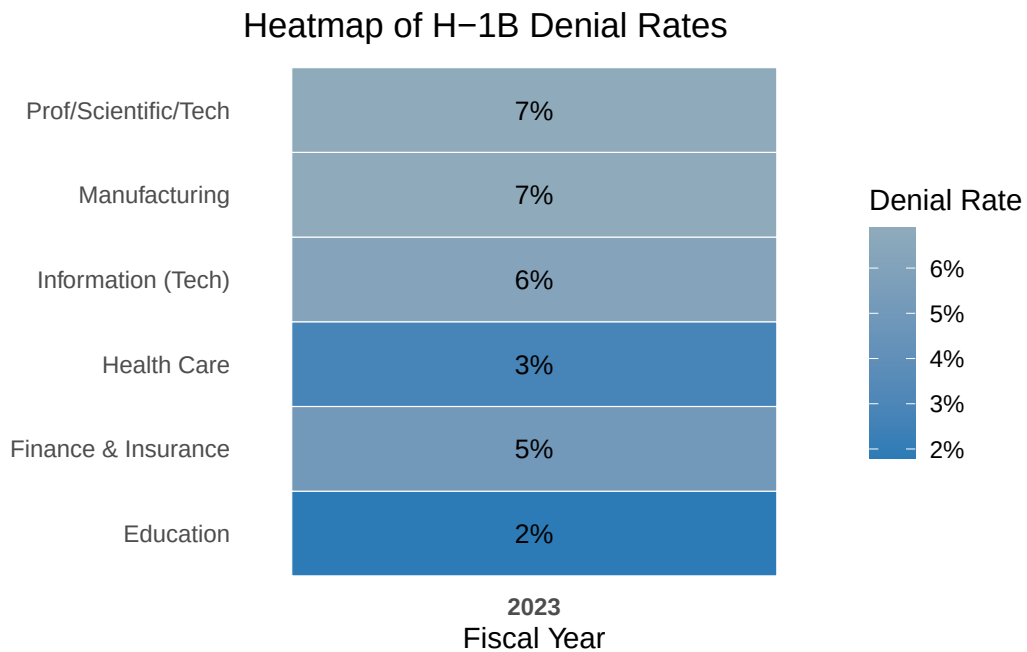
```
ggplot(industry_trends, aes(x = factor(fiscal_year), y = industry, fill = denial_rate))
  geom_tile(color = "white") +
  scale_fill_gradient2(low = "#2c7bb6", mid = "#ffffbf", high = "#d7191c",
    # set midpoint at 15% denial
    midpoint = 0.15,
    labels = percent_format(accuracy = 1)) +
  geom_text(aes(label = percent(denial_rate, accuracy = 1)),
    color = "black", size = 3.5) +

  labs(
    title = "Heatmap of H-1B Denial Rates",
```

```

x = "Fiscal Year",
y = NULL,
fill = "Denial Rate"
) +
theme_minimal() +
theme(
  panel.grid = element_blank(),
  axis.text.x = element_text(face = "bold")
)

```



```

ggsave("denial_rate_heatmap.png", width = 9, height = 6, bg = "white")

```

Facet Wrap Plot

```

ggplot(industry_trends, aes(x = fiscal_year, y = denial_rate)) +

  geom_rect(data = eras, inherit.aes = FALSE,

            aes(xmin = start, xmax = end, ymin = -Inf, ymax = Inf, fill = administration),

            alpha = 0.1) +

  geom_line(color = "#2c7bb6", linewidth = 1) +

  geom_area(fill = "#2c7bb6", alpha = 0.3) +

  # facet by Industry

  facet_wrap(~industry) +

  scale_y_continuous(labels = percent_format(accuracy = 1)) +

  scale_x_continuous(breaks = c(2017, 2020, 2023)) +

  scale_fill_manual(values = c("Trump" = "#d7191c", "Biden" = "#2c7bb6")) +

  labs(

    title = "Denial Rate Trends Broken Down by Key Industries (2017-2023)",

    x = "Fiscal Year",

    y = "Denial Rate"

  ) +

  theme_minimal() +

  theme(

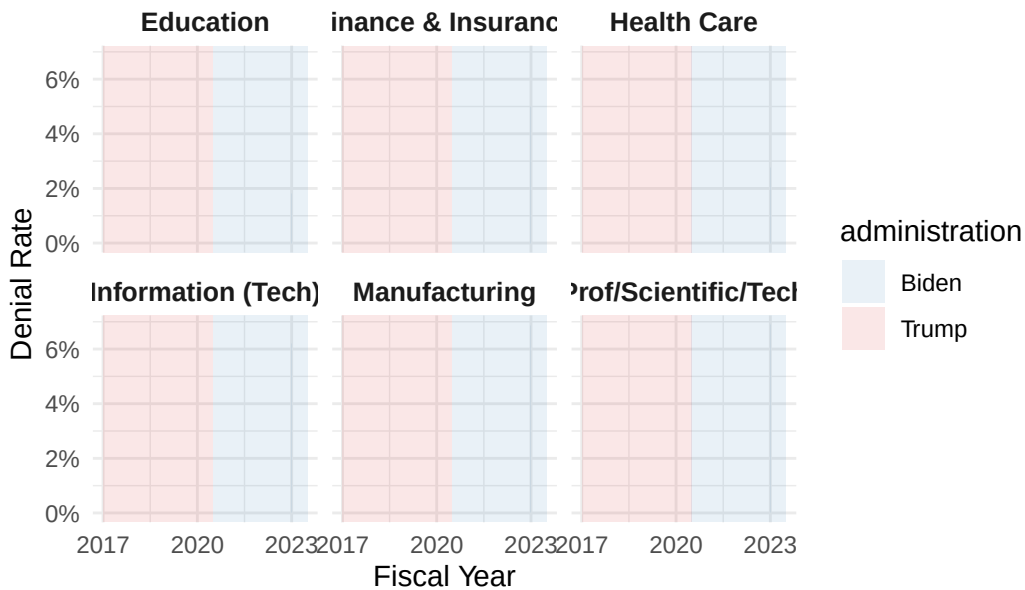
    strip.text = element_text(face = "bold", size = 10),

    legend.position = "right"

  )

```

Denial Rate Trends Broken Down by Key Industries (2017–2022)



```
ggsave("denial_rate_industry_facet.png", width = 12, height = 8, bg = "white")
```

Denial Rate by Company Size

```
size_analysis <- h1b_all_years |>
  group_by(employer) |>
  summarise(
    total_apps = sum(initial_approval + initial_denial, na.rm = TRUE),
    total_denials = sum(initial_denial, na.rm = TRUE)
  ) |>
  # create bins for company size
  mutate(size_category = case_when(
    total_apps <= 10 ~ "Small (1-10 apps)",
    total_apps <= 50 ~ "Medium (11-50 apps)",
    total_apps <= 500 ~ "Large (51-500 apps)",
```

```

    total_apps > 500 ~ "Mega (500+ apps)"

  )) |>

  # reorder the factors so they appear logically in the plot

  mutate(size_category = factor(size_category,

    levels = c("Small (1-10 apps)", "Medium (11-50 apps)",

      "Large (51-500 apps)", "Mega (500+ apps)"))))

size_rates <- size_analysis |>

  group_by(size_category) |>

  summarise(

    bucket_apps = sum(total_apps),

    bucket_denials = sum(total_denials)

  ) |>

  mutate(denial_rate = bucket_denials / bucket_apps)

ggplot(size_rates, aes(x = size_category, y = denial_rate, fill = size_category)) +

  geom_col() +

  scale_y_continuous(labels = percent_format(accuracy = 1)) +

  scale_fill_brewer(palette = "Blues") +

  labs(

    title = "Comparing Denial Rates By Company Size",

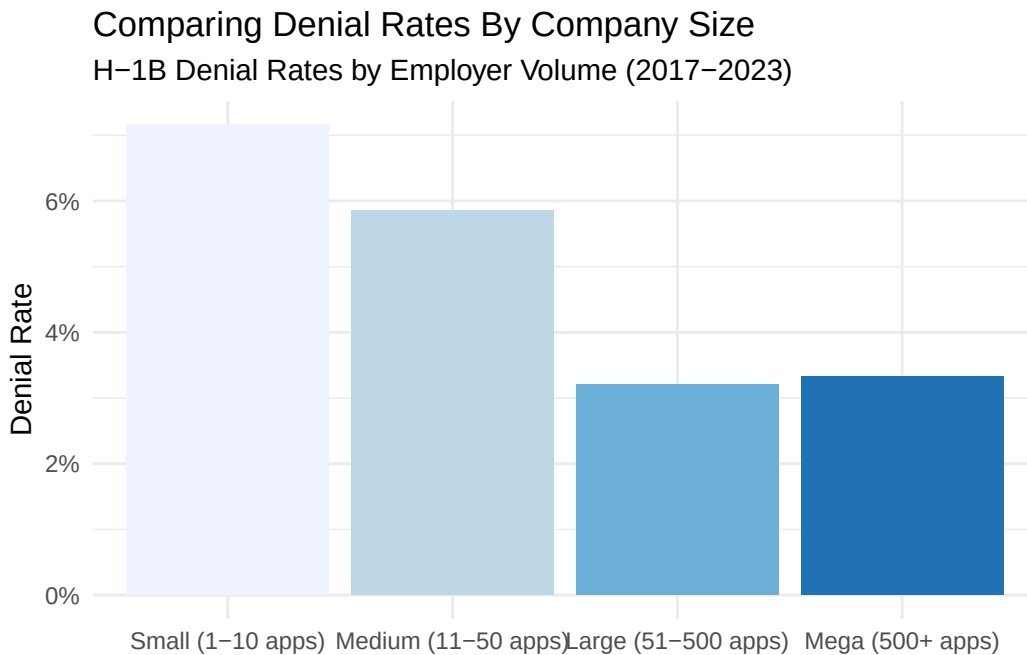
    subtitle = "H-1B Denial Rates by Employer Volume (2017-2023)",

    x = NULL,

    y = "Denial Rate"

```

```
) +  
  
theme_minimal() +  
  
theme(legend.position = "none")
```



```
ggsave("denial_rate_company_size.png", width = 8, height = 6, bg = "white")
```

US Map State Plot

```
state_risk <- h1b_all_years |>  
  group_by(state) |>  
  summarise(  
    total_apps = sum(initial_approval + initial_denial, na.rm = TRUE),  
    total_denials = sum(initial_denial, na.rm = TRUE)  
  ) |>  
  
# ignore states with tiny volume to avoid skewed 100% rates  
filter(total_apps > 1000) |>
```

```

mutate(denial_rate = total_denials / total_apps) |>

rename(state_abbr = state) |>

mutate(state = state_abbr)

plot_usmap(data = state_risk, values = "denial_rate") +

  scale_fill_distiller(

    palette = "Reds", direction = 1,

    name = "Denial Rate",

    label = scales::percent

  ) +

  labs(

    title = "Where is it Hardest to Get Approved?",

    subtitle = "Heatmap of Denial Rates by State (Min. 1,000 Apps, 2017-2023)"

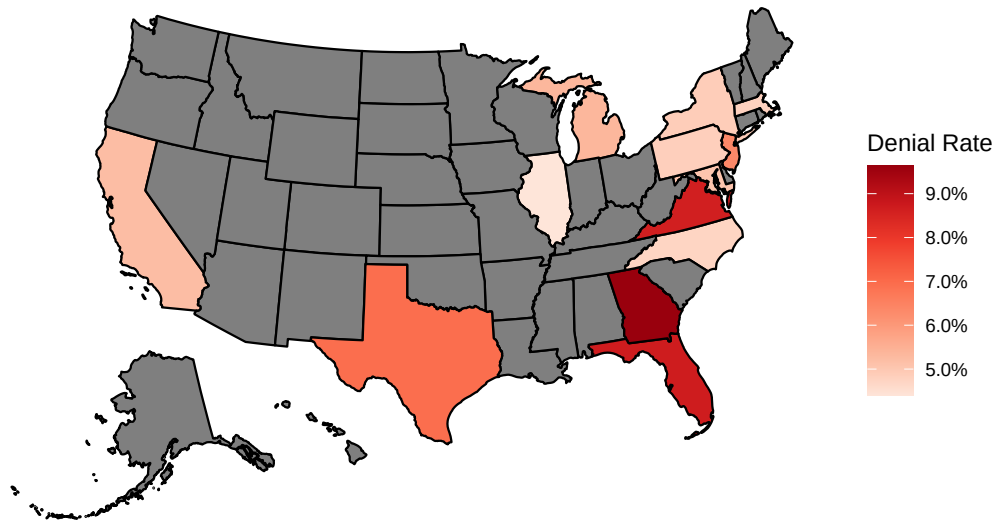
  ) +

  theme(legend.position = "right")

```

Where is it Hardest to Get Approved?

Heatmap of Denial Rates by State (Min. 1,000 Apps, 2017–2023)



```
ggsave("denial_rate_state_map.png", width = 10, height = 8, bg = "white")
```

New Hire Ratios

```
growth_analysis <- h1b_all_years |>

  mutate(naics_code = as.character(naics)) |>

  mutate(industry = naics_map[naics_code]) |>

  mutate(industry = ifelse(is.na(industry), "Other/Unknown", industry)) |>

  filter(industry %in% top_industries_list) |>

  group_by(industry) |>

  summarise(

    initial_total = sum(initial_approval + initial_denial, na.rm = TRUE),

    continuing_total = sum(continuing_approval + continuing_denial, na.rm = TRUE)

  ) |>

  mutate(
```

```

    total_activity = initial_total + continuing_total,

    new_blood_ratio = initial_total / total_activity

)

ggplot(growth_analysis, aes(x = total_activity, y = new_blood_ratio, label = industry))

  geom_point(aes(color = industry), size = 5) +

  geom_text_repel(size = 4) +

  # gog scale because Tech is huge

  scale_x_log10(labels = comma) +

  scale_y_continuous(labels = percent, limits = c(0, 0.6)) +

  labs(

    title = "Comparing Percentage of Newly Approved Petitions To Continuing Approvals",

    subtitle = "All Years (2017-2023)",

    x = "Total Activity (Log Scale)",

    y = "% of Petitions that are NEW Employment"

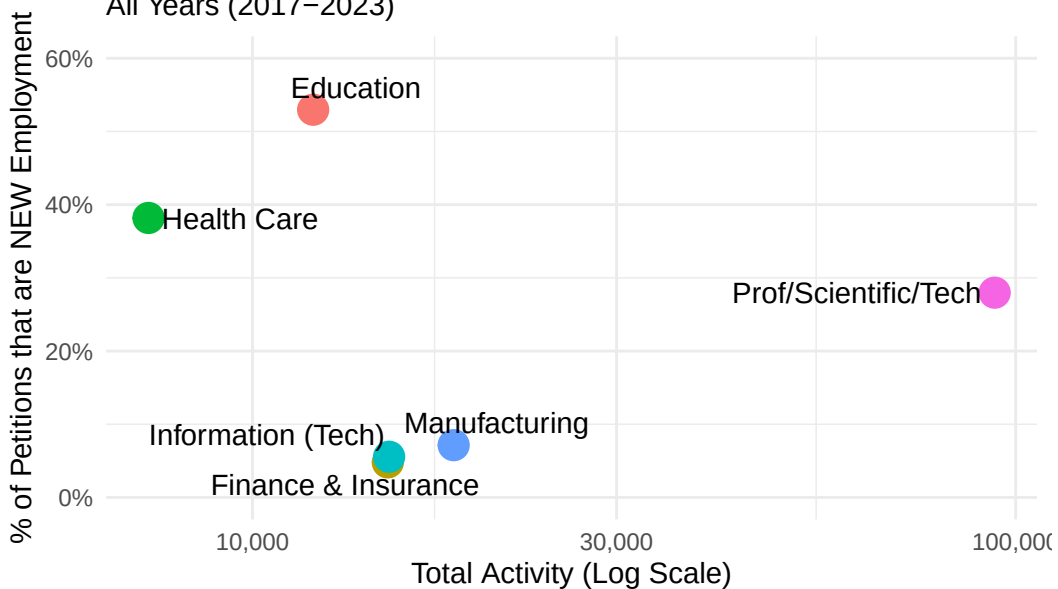
  ) +

  theme_minimal() +

  theme(legend.position = "none")

```

Comparing Percentage of Newly Approved Petitions To Continue All Years (2017–2023)



```
ggsave("new_hire_ratios.png", width = 8, height = 6, bg = "white")
```

Claudia

```
# Colors & Context

h1b_colors <- c(
  "Continuing Approval" = "#2c7bb6", "Initial Approval" = "#abd9e9",
  "Continuing Denial" = "#d7191c", "Initial Denial" = "#ff9896"
)

eras <- data.frame(
  administration = c("Trump", "Biden"),
  start = c(2017, 2020.5),
  end = c(2020.5, 2023.5)
)
```

```

# Plot: Petition Volume & Policy Landscape (2017-2023) ---

yearly_trends <- h1b_all_years %>%

  group_by(fiscal_year) %>%

  summarise(across(contains("approval") | contains("denial"), sum, na.rm = TRUE)) %>%

  pivot_longer(-fiscal_year, names_to = "status", values_to = "count") %>%

  mutate(status = str_replace_all(status, "_", " ") %>% str_to_title())

ggplot() +

  geom_rect(data = eras, aes(xmin = start, xmax = end, ymin = -Inf, ymax = Inf, fill = status),
            alpha = 0.1, inherit.aes = FALSE) +

  geom_col(data = yearly_trends, aes(x = fiscal_year, y = count, fill = status), width = 1) +

  scale_y_continuous(labels = comma) + scale_x_continuous(breaks = 2017:2023) +

  scale_fill_manual(values = c(h1b_colors, "Trump" = "#d7191c", "Biden" = "#2c7bb6"), blabel = "Case Status") +

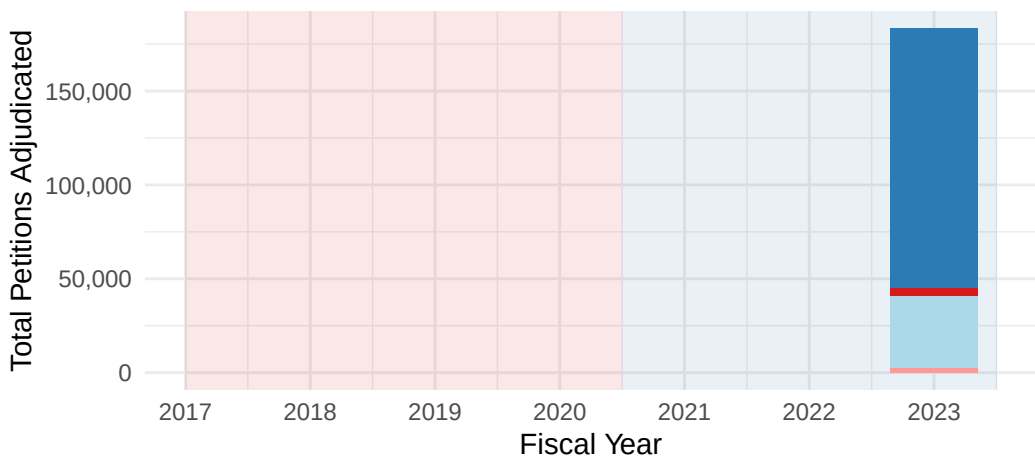
  labs(title = "H-1B Petition Volume & Policy Landscape (2017-2023)",
        subtitle = "Annual volume of adjudicated petitions across administrative shifts.",
        x = "Fiscal Year", y = "Total Petitions Adjudicated", fill = "Case Status") +

  theme_minimal() + theme(legend.position = "bottom", plot.title = element_text(face = "bold"))

```

H-1B Petition Volume & Policy Landscape (2017–2023)

Annual volume of adjudicated petitions across administrative shifts.



Case Status ■ Continuing Approval ■ Initial Approval ■ Continuing Denial ■ Ir

```
# Plot New Talent Acquisition Velocity by Industry

industry_trends <- h1b_all_years %>%

  mutate(industry = naics_map[as.character(naics)]) %>%

  mutate(industry = replace_na(industry, "Other/Unknown")) %>%

  group_by(fiscal_year, industry) %>%

  summarise(initial = sum(initial_approval + initial_denial, na.rm = TRUE),

            total = sum(initial_approval + initial_denial + continuing_approval + continuing_denial,
                        .groups = "drop") %>%

  mutate(new_hire_ratio = initial / total) %>%

  filter(industry %in% c("Professional, Scientific, & Technical", "Information",
                        "Manufacturing", "Finance & Insurance", "Educational Services",

ggplot(industry_trends, aes(x = fiscal_year, y = new_hire_ratio, color = industry)) +

  geom_line(linewidth = 1.2) + geom_point(size = 3) +
```

```
scale_y_continuous(labels = percent) + scale_x_continuous(breaks = 2017:2023) +
labs(title = "New Talent Acquisition Velocity by Industry",
      subtitle = "Initial petitions as a percentage of total adjudicated volume.",
      x = "Fiscal Year", y = "Initial-to-Total Petition Ratio", color = "Industry") +
theme_minimal() + theme(legend.position = "bottom")
```

